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MIL-T-23435F
13 May 1991
SUPERSEDING
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6 September 1985

MILITARY SPECIFICATION

TRUCK, TANK: SEWAGE OR WASTE COLLECTION, COMMERCIAL 4 BY 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial, four-wheel, two-rear-wheel-drive, diesel-engine-driven truck equipped with tank, and power-takeoff-driven vacuum pump, for collecting waste or raw unscreen sewage by suction.

1.2 Classification. Truck shall be of the following types, as specified (see 6.2):

Type I - 1,300 gallon minimum to 1,500 gallon maximum capacity pressure vessel.
Type II - 2,000 gallon minimum capacity pressure vessel.

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2320

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SPECIFICATION

MILITARY

MIL-V-62038 - Vehicle, Wheeled, Preparation For Shipment And Storage Of.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking& Data Plate Standards.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF ENERGY (DoE)

Title 10, CFR, Part 40 - Domestic Licensing of Source Material.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

* 2.2.1 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII, Division I- Pressure Vessels.

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(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

THE EUROPEAN TYRE AND RIM TECHNICAL ORGANIZATION (ETRTO)

Standards Manual.

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organization, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Standards and Recommended Practices

J534	- lubrication Fittings.
J537	- Storage Batteries (DoD adopted).
J551	- Performance Levels of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J688	- Truck Ability Prediction Procedure (DoD adopted).
J704	- Openings For Six-and Eight-Bolt Truck Transmission Mounted Power Take-offs.
J1349	- Engine Power Test Code- Spark Ignition and Diesel.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, Suite 304, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall consist of a pressure vessel, a power-takeoff-driven vacuum pump, complete with all necessary piping, valves, controls, and safety devices. The sewage collection components shall be mounted on a four-wheel, two-rear-wheel-drive, diesel-engine-driven chassis with conventional cab. Unit shall be DoT certified for hazardous, flammable, corrosive, class B poison or combination thereof.

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3.2 First production vehicle. The contractor shall furnish a truck for first production vehicle inspection.

3.3 Standard truck, components, and accessories. Except as specified in 3.3.1 through 3.3.9, the truck, components, and accessories shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to the authorized company representative for selection of truck models and components. Truck components, assemblies, and accessories to be provided hereunder are standard or optional commercial products for the purpose of this specification. Such items shall be construed as commercial motor vehicle products, and shall comply with all DoT Federal Motor Vehicle Safety Standards and Regulations applicable, on the date of manufacture.

* 3.3.1 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished which shall maintain coolant at a temperature below the boiling point, with truck loaded to its gross vehicle weight rating (gvwr) and operated at 10,000 feet above sea level, or in an ambient air temperature of not less than 125 degrees Fahrenheit (^{o}F).

3.3.2 Drain plugs. Permanent magnet type drain plugs shall be provided for the manual transmission and the rear axle.

3.3.3 Coolant system and indicators. Coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than 2 quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the cab dash instrument panel.

3.3.3.1 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided and shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

3.3.4 Towing devices. Not less than two hooks or loops, mounted to the frame or to rigid members which are attached to the frame, shall be furnished on the front of the truck for towing purposes.

3.3.5 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.3.6 Splash guards. The truck shall be equipped with manufacturer's standard splash guards. Splash and stone throw protection shall be in accordance with SAE J682.

* 3.3.7 Sound level. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks. The interior sound level shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.94.

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* 3.3.8 Air pollution control. The truck shall comply with the Environmental Protection Agency (EPA) Regulations governing Control of Air Pollution from New Motor Vehicles Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with the State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.9 Safety. The truck shall not contain any radioactive material. Radioactive material is defined as that material in which the activity is in excess of 0.002 microcuries or greater as identified in Title 10, CFR, Part 40 also no part of the truck shall contain any asbestos material.

3.4 Identical items. All units of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.5 Vehicle weights and ratings.

3.5.1 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricants, and coolant.

* 3.5.2 Weight rating. The gross vehicle weight rating (gvwr) shall consist of curb weight, operator (weight computed at 175 pounds [lbs]), and a payload computed at 8.3 lbs per gallon to provide not less than the specified gvwr.

3.5.3 Weight distribution. The distribution of the gvwr for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

* 3.5.4 Ratings. Component and chassis ratings shall not be raised to meet the requirements of this specification. Minimum gvwr shall be in accordance with Table I.

Table I. Minimum gross vehicle weight rating.

Type	GVWR (lbs)
I	24,000
II	34,000

3.5.5 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors, and tires, shall be not more than 96 inches.

3.6 Cab-to-axle. The cab to axle dimension shall be in accordance with the body manufacturer's recommendation.

3.7 Performance.

3.7.1 Speeds and gradeability. High and low speed requirements shall be met with truck loaded to specified gvwr.

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3.7.1.1 High speed gradeability. Truck shall ascend a continuous 1.0 percent grade at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and with the multispeed axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.4).

* 3.7.1.2 Low speed. Low speed, with engine operating at not less than 35 percent of recommended governed speed, shall provide a truck speed of more than 5 mph. The truck, fully loaded, shall be capable of starting from a parked location, facing uphill, on a grade of not less than 25 percent.

* 3.7.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 60 mph.

3.7.2 Brake performance. Service brakes shall comply with the performance requirements specified in DoT Federal Motor Carrier Safety Regulations, section 393.52. Service brakes shall control and hold the truck, loaded to the specified gvwr, on a 30 percent grade. Parking brake shall hold the truck loaded to the specified gvwr, headed either up or down the grade, without slipping on a 30 percent grade.

3.8 Chassis components.

3.8.1 Engine. The engine furnished shall be the chassis manufacturer's standard, or optional engine for the commercial model truck which meets or exceeds the requirements of this specification. The truck shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine with not less than four cylinders. Engine net horsepower (hp) figures used in performance prediction calculations shall be determined in accordance with SAE J1349. A fan clutch to automatically reduce fan speed, when not required for engine cooling, shall be provided.

3.8.1.1 Oil filter. A full flow type oil filter shall be furnished.

3.8.1.2 Governor. An engine governor shall be furnished, set, and sealed to limit engine to engine manufacturer maximum recommended operating speed.

3.8.2 Electrical system. The electrical system shall be in accordance with DoT Federal Motor Carrier Safety Regulations, sections 393.27 through 393.31 and 393.33.

3.8.2.1 Starting system. A 12 or 24-volt (V) starting system, with 12V direct current lighting system, and not less than 60 ampere (A) alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type ether systems shall be key operated or manually operated from the driver's compartment, shall have a reservoir or provisions for a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

* 3.8.2.2 Batteries. Each battery shall be of 12V potential. The total reserve capacity rating and the total cold cranking rating, measured in accordance with SAE J537, shall be not less than 320 minutes and 1200A respectively.

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* 3.8.2.3 Lighting. All truck lights, reflectors, and wiring shall be as specified herein, and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19 through 393.26, as applicable. Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected), or bumpers. All wiring shall be installed in vapor resistant conduit from the cab rearward. Brake lights shall override the emergency flashers of this truck.

3.8.3 Fuel system. The fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.65 and 393.67.

* 3.8.3.1 Air cleaner. A dry type air cleaner meeting engine manufacturer's requirements shall be furnished.

* 3.8.3.2 Fuel tank(s). The fuel tank(s) shall be the manufacturer's standard for the truck furnished, except that the total capacity shall be not less than 40 gallons. When more than one fuel tank is furnished, means shall be provided to assure equalized fuel level in both tanks.

* 3.8.4 Exhaust system. The exhaust system shall conform to DoT Federal Motor Carrier Safety Regulation, section 393.83. Vertical exhaust mufflers capable of being reached easily by personnel entering or leaving either side of the cab shall be provided with a heat shield. Vertical exhaust systems shall be provided with a hinged rain cap.

3.8.5 Transmission. The truck shall have a manually shifted transmission. Transmission shall have not less than five forward speeds. Input torque capacity of transmission shall be at least equal to maximum torque delivered by the engine. Transmission and axle gear ratio shall be selected to provide performance specified in 3.7.1 Transmission shall be provided with SAE J704 power takeoff opening on both right and left sides.

3.8.5.1 Clutch. The clutch shall be the largest capacity clutch offered for the truck and engine furnished, with torque capacity exceeding maximum delivered engine torque.

3.8.5.2 Drive line components. The drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.8.5.3 Power takeoff drive. A power takeoff shall be furnished to operate the vacuum pump and shall be heavy duty and continuous operation type. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.8.6 Frame. The chassis frame shall be manufacturer's standard for the truck furnished.

* 3.8.7 Suspension. The truck shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the truck loaded to specified minimum gvwr, (see 3.5.2 and 3.5.4). When suspension capacity is rated at the spring pads,

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the unsprung weight shall be deducted. The truck shall be equipped with hydraulic, double-acting, shock absorbers at the front wheels.

* 3.8.8 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the truck loaded to specified minimum gvwr. Two-speed rear axle shall be furnished, equipped with electric-, or air-shift and provided with ratios which will permit proper gear splitting. Rear axle gear ratio shall provide performance specified in 3.7.1.

* 3.8.9 Wheels, rims, tires, and tubes. The truck shall be equipped with single front and dual rear wheels. Rim and tire ratings shall conform to TRA or European Tyre and Rim Technical Organization recommendations, for the type and size tires furnished. Tire and rim sizes shall be the same for all wheels on each truck. Disk-type wheels shall be furnished.

* 3.8.9.1 Tires. The tires shall be tube or tubeless, truck type radial ply, with highway tread. The tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with truck loaded to rated gvwr. Tires shall conform to TRA or to The European Tyre and Rim Technical Organization recommendations, with the same size designation.

* 3.8.9.2 Tubes. When tube-type tires are furnished, tubes shall be of heavy-duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube-type tires in accordance with TRA or The European Tyre and Rim Association recommendations.

3.8.9.3 Spare wheel assembly carrier. A carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the truck.

3.8.9.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be provided and installed on the carrier.

3.8.9.5 Spare tire. When specified (see 6.2), an inflated spare tire mounted on the spare wheel or rim shall be furnished. The spare tire shall be of the same size and tread design and of the highest ply rating furnished on the truck.

3.8.9.6 Tire chain clearance. There shall be no interference that would prevent attaching and using chains on rear wheels.

3.8.10 Brakes. The brakes shall conform to Dot Federal Motor Carrier Safety Regulations, sections 393.40 through 393.43 and 393.45 through 393.52, and as specified herein.

* 3.8.10.1 Service brakes. Truck shall be equipped with full-air service brakes. Full-air braking system, complete with all necessary components, including, but not limited to:

a. Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7 1/4 cubic feet per minute (cfm).
b. Air storage reservoir, with not less than 2,000 cubic inch total capacity, each tank equipped with drain, and with safety and check

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valves between compressor and first reservoir tank.
c. Foot pedal control.
d. Air control valves.
e. Air pressure gage, visible to the driver.
f. Low-air-pressure warning, visible and audible.
g. Alcohol aspirator, with unbreakable transparent container.
h. Automatic moisture ejector.

3.8.10.2 Air brake emergency system. An emergency air brake system shall be provided.

* 3.8.10.3 Parking brake. The truck shall be equipped with a manually actuated parking brake.

3.8.10.4 Brake control from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front on a protected position providing for a ready attachment of air hose from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any DoT Federal Motor Carrier Safety Regulations referenced herein or to any DoT Federal Motor Vehicle Safety Standard.

* 3.8.11 Cab. The cab shall be full width providing a minimum of two seating positions, one each for the operator and one passenger. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least one door equipped with external, key-operated lock. Drip rails shall be installed above the cab doors. Cab shall have upholstered, full-width, adjustable seat and back or individual adjustable driver's seat and individual passenger seat. The color of the upholstery and the interior finish shall be compatible with the exterior color. White upholstery shall not be furnished. Safety grips shall be provided on each side of the cab to assist personnel to enter or leave the cab. Interior dome lighting shall be provided. Seats belts shall be furnished for each seating position in the cab.

3.8.12 Steering. Manufacturer's standard power steering shall be furnished.

3.8.13 Windshield wipers and washers. The truck shall be equipped with dual windshield wipers and windshield washers. The windshield wipers shall be multispeed type and operated by either air or electric motor(s).

3.8.14 Front bumper. Channel type front bumper shall be furnished.

3.8.15 Rear end protection. Rear end of the truck shall be protected in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.86. Rear bumper shall be furnished.

3.8.16 Tools. When specified (see 6.2), the truck shall be furnished with tools required for exchanging mounted tire assembly with a spare wheel and tire assembly, and shall include as a minimum a hydraulic jack, jack handle, and

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wheel nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising the truck while fully loaded, to a height adequate to permit removal and replacement of wheel and tire assembly.

3.8.17 Heater and defroster. A hot-water heater shall be provided. The heater shall have fresh air intakes. Discharging outlets shall be provided to direct heated air to the flow and to defroster louvers. The heater shall be complete with blower and shall have mounted controls convenient to the driver.

3.8.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be permanently identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be panel mounted.

3.8.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Key-operated ignition switch.
b. Ammeter, charging indicator, or voltmeter.
c. Fuel gage.
d. Oil pressure gage or indicator.
e. Engine temperature gage or indicator.
f. Speedometer with recording odometer.
g. Dual sunvisors.
h. Door mounted armrest on driver's side.
i. Cab ventilator other than windows.
j. Manufacturer's standard electric horn.
k. Tachometer.

* 3.8.20 Rearview mirrors. Outside rearview mirrors, each with at least three supporting arms, shall be mounted on each side of the cab. Both flat and convex mirror heads (may be combined assembly) shall be furnished. The mirror surfaces shall be not less than 50 square inches in the flat heads, and 20 square inches in the convex heads.

3.8.21 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.9 Sewage collection equipment. Sewage contents shall be lifted into the tank through a hose line by suction, provided by the partial vacuum developed within tank. Unloading of sewage from truck shall be gravity and by compressed air to evacuate the tank of sewage through rear gate valve. The required partial vacuum and air pressure shall be generated by a dual-purpose pump

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furnished with the truck and designated herein as the vacuum pump. Vacuum pump shall be driven by power takeoff from the truck engine.

3.9.1 Tank and vacuum pump performance. The vacuum pump shall be of adequate capacity to provide an average water flow rate into the sewage tank of not less than 300 gallons per minute (gpm) with a vertical lift of 10 feet; an average water flow rate of not less than 200 gpm with a vertical lift of 18 feet; and an average water flow rate of not less than 20 gpm with a vertical lift of 25 feet. Evacuation of sewage from the tank shall be at an average water flow rate of not less than 250 gpm when using compressed air from the discharge side of the vacuum pump. All sewage inlet and outlet piping shall be capable of passing solids up to 2-inch sphere size. Float shutoff valves of sewage tank and surge tank shall be capable of closing vacuum line, when under suction from vacuum pump, without transmitting any liquid to vacuum pump when either the sewage tank or surge tank is filled with water. The sewage tank valve and surge tank shall be capable of preventing sewage foam from entering the vacuum pump suction line when truck is loading waste caustics and detergents, to within an allowable leakage tolerance of 1/50 of 1 percent of the sewage tank capacity collected into the tank.

* 3.9.2 Sewage tank. The sewage collection tank shall meet capacities listed, see Table II. The sewage tank shall be horizontal type, in accordance with ASME Boiler and Pressure Vessel Code, section VIII, for a design pressure of not less than 40 pounds per square inch and for full vacuum. ASME Boiler and Pressure Vessel Code, section VIII, certification shall be furnished by an authorized Boiler and Vessel Inspector. The tank shall be fabricated from not less than 1/4-inch, high tensile, low alloy steel; or from mild steel or copper bearing steel of a thickness that provides equivalent strength, with welded smooth skin construction. The tank heads shall be substantially dished, flanged, and reinforced to provide structural strength of hydraulic pressure from either direction. The length of the tank shall be such as to give load distribution proportional to the rated capacity of the front and rear axles of the truck chassis. Outlet opening shall be at rear bottom of the tank located to provide complete tank drainage. Full length longitudinal Z-type, box type, or V-type tank frame members shall be furnished.

Table II. Tank Capacity in Gallons

Type	Minimum	/	Maximum gallons
I	1,300	/	1,500
II	2,000	/	2,300

3.9.2.1 Baffles. Tank shall be provided with baffles, the number of which shall be such that the linear distance between any two adjacent baffles, or between tank head and the baffle nearest to it, shall not exceed 50 inches. Baffles shall be reinforced and provided with half round openings at top and bottom. Each baffle shall be provided with manway opening not less than 16 inches in diameter. The baffle area shall be not less than 80 percent of the cross-sectional area of the tank.

3.9.2.2 Access openings. Top of tank shall be furnished with an access opening located to provide accessibility to the tank float shutoff valve. In addition, a washout opening shall be furnished on the top of the tank. Each opening shall be not less than 16 inches in diameter, or an oval with axis of

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not less than 12 by 16 inches. Each opening shall be provided with a cover. Each cover shall be provided with synthetic rubber gasket, and shall be of the quick-opening type capable of being removed without the use of special tools.

* 3.9.2.3 Sewage intake. Sewage intake shall be from rear top of the tank, or from rear bottom through a vertical standpipe inside the tank extending up within 18 inches of tank top. Standpipe shall be provided with a baffle wear plate. Standpipe shall be not less than 3-inch inside diameter (id).

3.9.2.4 Tank sewage safety valve. A sewage tank safety valve shall be provided. The pressure relieving safety valve shall be of copper-alloy and shall be sized and installed in accordance with requirements of ASME Boiler and Pressure Vessel Code, section VIII. The safety valve shall have a relieving capacity not less than the maximum output capacity of the vacuum pump.

3.9.2.5 Surge tank. A surge tank of not less than 1-gallon capacity shall be furnished in the vacuum line between sewage tank and vacuum pump. Means shall be provided to drain and clean surge tank.

3.9.2.6 Accessories and valves. Accessories and valves furnished with the sewage tank shall include at least the following:

a. A cleanout opening of not less than 75 square inch size at the bottom rear end of the tank, provided with quick-opening cover and rubber gasket.
b. Gage(s) with not less than 3-inch diameter dial(s) to indicate the partial vacuum attained when loading tank, the air pressure during the time tank is loaded with sewage, and when evacuating tank.
c. Mechanical liquid level indicator graduated in quarters of tank capacity.
d. Quick opening inlet and outlet sewage gate valves of not less than 3-inch and 4-inch nominal size, respectively, fitted with cam-locking type hose couplers and furnished with blank dust caps for the couplers.

All valves and drain cocks shall be at least the commercial quality of iron body valves with corrosion-resistant seats. All shutoff valve floats shall be type 316 stainless steel, copper, copper-alloy, and a minimum of 8 inches in diameter.

3.9.2.7 Sewage tank mounting. Sewage tank subframe shall be secured with U-bolts, twin studs, or by brackets and the mounting shall include a hardwood breaker strip.

3.9.2.7.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than three U-bolts or twin studs, per side, each having 9/16-inch minimum body diameter with 5/8-inch minimum thread diameter. Tie plates shall be at least 1/2-inch thick. Truck chassis frame shall be braced, using hardwood blocks at each mounting point unless mounting point is located at a full depth, frame crossmember. Blocks shall incorporate a keeper strap or groove for mounting bolt, and shall be of a width to assure retention. Two shear bolts shall be provided, one on each side of rear portion of body subframe, to maintain body alinement on truck chassis. Forward mounting bolts

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shall be located to the rear of the tapered portion of breaker strips (see 3.9.2.7.3).

3.9.2.7.2 Brackets. When brackets are used, they shall be bolted to the web of the chassis frame rails. The mounting brackets shall provide means of drawing down the body subframe on the chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure mounting brackets to chassis frame rails, they must be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

3.9.2.7.3 Breaker strips. A hardwood breaker strip, not less than 1 1/6 inches finished thickness, shall be installed between sewage tank longitudinal sills and truck chassis frame. Breaker strips shall have a taper of 1 inch in 18 inches at the forward end.

* 3.9.3 Piping. Unless otherwise specified (see 6.2), all vacuum lines inside the tank shall be steel or iron pipe with corrosion-resistant fittings. The pipe and fittings shall be not less than 2 inches standard iron pipe sizes (ips), and shall be made up in subassemblies and interconnected by boiler unions or similar quick disconnects. Vacuum line piping outside tank shall be not less than 2 inches ips, of steel or hard copper pipe with matching fittings of iron or copper-alloy, respectively. Flexible hard rubber hose is acceptable.

* 3.9.4 Vacuum pump. The vacuum pump shall be water- or air-cooled, multicylinder, reciprocating piston pump or rotary type pump with automatic oilers, capable of the performance required in 3.9.1 for both partial vacuum and compressed air output, and shall be furnished complete with all necessary accessories. The vacuum pump suction and discharge openings shall be minimum of 2 inch id. The vacuum pump shall have a continuous duty pressure of 150 cfm at 26 inches of mercury, and a minimum discharge pressure of 35 lbs per square inch. Vacuum pump mounting shall be such as to prevent undue vibration. Ground clearance for the vacuum pump shall be not less than 12 inches. A four-way valve, or four valves, shall be provided on vacuum pump manifold in order to permit the tank supply line to be switched between the partial vacuum and the compressed air outlets of the vacuum pump, control(s) for the operator shall located at the side or rear of truck. Each or all valves shall be operated from a control lever capable of remaining in any position setting unaffected by vibration. Exhaust air from the vacuum pump shall be piped to an outlet located on the opposite side of truck from the truck engine tailpipe.

3.9.5 Hoses and hose carriers. Unless otherwise specified (see 6.2), eight 25-foot lengths of suction hose, and four 25-foot lengths of discharge hose shall be furnished. Hose shall be smooth bore, rated at full vacuum and pressure of not less than 45 pound-force per square inch (psi). Each section shall be fitted with male and female copper-alloy or aluminum couplings of the cam-lock type, rated for not less than 100 psi. Troughs shall be provided for stowing the hose sections and shall be provided with sufficient drainage. Hose sections shall be attached to the troughs with hose clamps.

3.9.6 Storage compartments. The truck shall be equipped with an underslung compartment on each side of the sewage tank. Compartments shall have hinged doors with locks operable with the same key.

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3.9.7 Walkway and access ladder. The truck shall be equipped with a walkway on the curbside of the sewage tank. The walkway shall extend the full length of the sewage tank and shall have nonskid surface. The walkway shall provide an unobstructed surface. An access ladder with grab rails or grab handles, located on the curbside of the truck, shall be provided to gain access from the walkway to the sewage tank access openings. The ladder steps and grab handles shall be provided to gain access from ground level to the walkway.

3.9.8 Rear wheel fenders. Fenders shall be furnished and mounted to cover rear wheels for an arc of not less than 135°. Fenders shall be of one-piece construction.

* 3.9.9 Instruction plate. A metal plate, showing a schematic layout of all valves, valve controls, and piping system with full operating instructions, shall be fastened on the inside of curbside compartment door. The metal plates and markings shall be constructed so as to last not less than seven years while exposed to the environment elements.

3.9.10 No smoking signs. Metal signs stating "NO SMOKING WITHIN 50 FEET", in painted enamel lettering not less than 3 inches high, shall be permanently fastened on both sides and rear of truck without interfering with the truck markings specified (see 3.11.1).

3.9.11 Floodlight. When specified (see 6.2), a sealed unit floodlight shall be furnished and mounted in a swivel type holder at the rear of the truck. The floodlight shall have a weatherproof switch accessible from the ground.

3.10 Lubricating. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.11 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

3.11.1 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223.

3.12 Rustproofing. When specified (see 6.2), the truck cab, chassis, and chassis underside shall be rustproofed in accordance with FED-STD-297.

3.13 Servicing and adjusting. Prior to acceptance of the truck by the Government, the contractor shall service and adjust the truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes, (when required).
c. Proper functioning of all lighting and electrical systems.

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d. Wheel alinement, (when required).
e. Adjustment of engine to include tune-up, (when required).
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

13.14.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

13.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

13.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system

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or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the test of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First production vehicle test.

4.4.1 Vehicle weight. The truck shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight, the operator weight at 175 lbs, and the payload (see 3.5.2), to provide the specified gvwr. The calculated imposed loads on front and rear axle will be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet the requirements of 3.8.6, 3.8.8, and 3.8.9.1.

* 4.4.2 Road test. The truck, loaded to the rated gvwr, shall be driven at maximum speeds limits (not less than 50 mph) for not less than 50 miles over

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improved surface roads, and not less than 20 mph over unimproved roads surfaces for 20 miles. Upon completion of the road tests, the vehicle shall be inspected for leaks, damage, or failure of components. Damage or failure of safety items, holding tank, and related items shall constitute failure of the test and cause rejection of unit.

4.4.3 Pumping equipment tests. Truck pumping equipment shall be tested, substituting water for sewage, to verify conformance to 3.9.1, as follows:

a. Test the vacuum pump for each average water flow rate with respective vertical lift. Each average water flow rate requirements shall be performed for not less than one rated sewage tank capacity.
b. Test for tank vacuation by air pressure from vacuum pump shall be performed for a period of not less than 5 minutes.
c. Test for tank float shutoff valve operation, under partial vacuum when water level reaches maximum tank capacity, shall be made with pump engine operating above half throttle.
d. Test for surge tank operation under partial vacuum shall be made with a rising water level to operate float shutoff valve, with pump engine operating half throttle.
e. Test for surge tank and float valves for leaks.
f. Test for passing 2-inch sphere sizes shall be made on both inlet and outlet piping system.
g. Test tank for leaks.

4.4.4 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as the production sample, and be the last truck shipped on the contract. The first production vehicle shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first production vehicle in a serviceable condition for the duration of the contract.

4.4.6 Body treatment and painting. The certification regarding the cleaning, treating, and prime painting, as required by MIL-STD-1223, shall be presented to government representatives for examination and approval.

4.4.7 Brake test. The truck loaded to its GVWR shall be placed in a grade not less than 30 percent for a time period of 1 hour to verify the brake performance specified in 3.7.2.

* 4.5 Production truck tests. The contractor's testing system shall, as a minimum, assure that the truck is capable of meeting the performance requirements specified in section 3. Each truck shall be tested for leaks. At the point of final acceptance the contractor shall make available to the government records acceptable to the government indicating that the servicing and adjusting required by 3.13 has been accomplished.

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4.6 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packed in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved, packed, and marked in accordance with the mobile requirements of MIL-V-62038 with the level of preservation and packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The trucks covered by this specification are intended for collection of sewage or waste.

* 6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type of truck required (see 1.2).
d. When first article inspection and approval is required (see 3.2).
e. When spare wheel or rim is required (see 3.8.9.4).
f. When spare tire is required (see 3.8.9.5).
g. When brake control from a towing vehicle is required (see 3.8.10.4).
h. When tire servicing tools are required (see 3.8.16).
i. When piping material other than as specified is required (see 3.9.3).
j. When hose section lengths other than as specified are required (see 3.9.5).
k. When floodlight is required (see 3.9.11).
l. When finish color coat shall be as specified (see 3.11).
m. When marking and data plates are required (see 3.11.1).
n. When rustproofing is required (see 3.12).
o. When preservation-packing in accordance with MIL-V-620238 is required, and the level of preservation-packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for speed requirements for each vehicle model furnished under contract should be submitted as specified in the

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contract. The SAE Work Sheet Item should include vehicle model number, wheelbase, engine model number, and vehicle type. Unless other conditions are sighted in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature and still, dry air. The factors to be used in predicting truck ability (see 3.7.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.	
Table 2	- Altitude factor.	1.00
Table 3	- Rolling factor.	1.613
Table 4	- Area factor. Use factor.	0.173
Table 5	- Velocity factor.	250.0
Table 6	- Altitude factor.	1.00

Table 7 - Chassis friction hp. Use applicable power unit gvw (to nearest, higher 1,000 lbs) and the engine revolutions per minute (to nearest 100 revolutions) which is required for 50 mph geared speed. For gvw and engine speed beyond the range of this table, factors shall be extrapolated.

Table 8	- Grade factor.	0.75
Table 8A	- Correction factor.	Not required
Table 9	- Road factor.	0.0

* 6.5 Subject term (key word) listing.

Engine
Pumping
Waste disposal

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Preparing Activity:
Navy - YD

(Project 2320-0455)

Review Activity:
Air Force - 84

User Activity:
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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-23435F	91-05-13

3. DOCUMENT TITLE

TRUCK, TANK: SEWAGE OR WASTE COLLECTION, COMMERCIAL 4 By 2

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER (156) NAVAL CONSTRUCTION BATTALION CENTER	805-982-5572	551-5572
c. ADDRESS (Include Zip Code) PORT HUENEME, CALIF. 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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